

Multiverse (Main UI)

3D orchestration platform with Three.js, 9 visual spaces, and voice dialog routing.

Overview

Multiverse is the main UI and orchestration hub of Vibemind. It renders 9 visual spaces as 3D objects with Three.js and offers voice dialog as the primary interaction mode. The voice interface routes user intention via IntentClassifier to the appropriate space agent — each space is independent and handles execution on its own. The voice agent only passes the user's intent to the correct agent.

3D Multiverse: 9 Visual Spaces

Defined in `electron-app/renderer/multiverse.js`:

Space	Icon	Agent Name	Role	Position	Color
ideas	■	Rachel	Idea Navigator	(0, 0, 0)	0x4488ff (Blue)
projects	■	Coding	Project Manager	(16, 0, 11)	0x44ff88 (Green)
desktop	■	Hassan	Desktop Worker	(22, 0, -14)	0xff8844 (Orange)
roarboot	■	Rowboat	Knowledge Navigator	(-17, 0, -8)	0x22ccaa (Teal)
swedesign	■	Factory	Spec Generator	(8, 0, 5.5)	0xff6633 (Red-Orange)
clawport	■	Dashboard	System Dashboard	(-8, 0, 12)	0x8866ff (Purple)
agentfarm	■	Farmer	Agent Orchestrator	(-14, 0, 18)	0x88aa44 (Olive)
thebrain	■	Brain	Knowledge Center	(14, 0, -22)	0xff66aa (Pink)
video	■	Director	Video Producer	(-20, 0, 24)	0xee4466 (Red)

Note: Agent names are UI labels for the respective backend agents. Each space is independent and acts on its own — the voice agent (Rachel) merely routes the user's intention to the correct space agent. The spaces then handle complete execution independently.

Key Components

File	Purpose
<code>renderer/multiverse.js</code>	MultiverseApp: Three.js scene, camera, Simplex noise, space navigation
<code>renderer/glass_bubbles.js</code>	Bootstrap, Roarboot status patching

File	Purpose
renderer/universe_canvas.js	Canvas rendering
renderer/exploration_dialog.js	Exploration UI
renderer/rich_content_renderer.js	Rich content display
renderer/shuttle_manager.js	Shuttle/requirements pipeline UI
renderer/lib/three.min.js	Three.js library
renderer/lib/OrbitControls.js	Camera control

3D Features

- **Simplex Noise Vertex Displacement:** Organic surfaces for space objects
- **Pulse Animation:** Lively space representation
- **Camera Interpolation:** Smooth navigation between spaces
- **DNA Helix Bubble Rendering:** Ideas as DNA helix structure
- **Hand Gesture Navigation:** Gesture-based control
- **Organic Brain Mesh:** 60 neuron particles with synaptic lines (TheBrain)

Technology Stack

- **3D Engine:** Three.js with OrbitControls
- **Voice Engine:** OpenAI Realtime API (speech-to-speech) via VoiceBridgeV2
- **Routing Engine:** IntentClassifier → EventRouter → Backend Agents
- **Event Bus:** python/swarm/event_bus.py (Pub/Sub for cross-space communication)
- **State Synchronization:** SQLite + IPC messages (Python ↔ Electron)
- **ClawPort Dashboard:** React-based monitoring overlay (5 tabs: schedule, agents, chat, memory, plugins)

Current Status

- 3D Multiverse with 9 visual spaces fully functional
- Voice dialog via Rachel → IntentClassifier → Backend Agents operational
- Three.js rendering with Simplex noise and animations
- Camera navigation between spaces working
- Shuttle manager for requirements pipeline integrated
- ClawPort React dashboard (schedule, agents, chat, memory, plugins)
- Exploration dialog UI for AI-scientist sessions
- Intent routing to 10 backend agents (all lazy-loaded in __init__.py)

Roadmap

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- Complete Desktop.Space context integration for full awareness (Q1-Q2 2026)
 - Implement advanced intent understanding with multi-intent detection
 - Build comprehensive event interlock system for complex workflows
 - Add real-time space performance monitoring and load balancing
 - Create workflow recording and replay for repeatable multi-space tasks
 - Implement learning system that improves routing over time
 - Add collaborative features for team-based workflows

Ecosystem Fit

Multiverse is the central nervous system of Vibemind. Every user interaction runs through Multiverse, which intelligently routes it to appropriate spaces. It coordinates Ideas.Space, Coding.Space, Desktop.Space, N8n.Space, Rowboat.Space, The Brain.Space, Video.Space, and Software Design.Space into a unified platform.